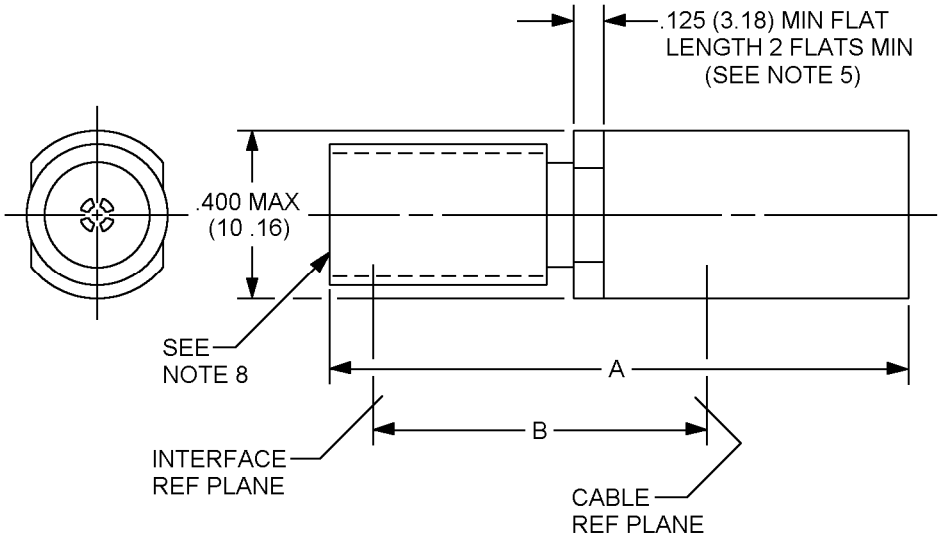
INCH-POUND
MIL-PRF-39012/81G
18 January 2007
SUPERSEDING
MIL-PRF-39012/81F
24 January 1992

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (CABLED, SEMIRIGED) – SOCKET CONTACT, CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39012.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Metric equivalents are in parentheses.
4. Receptacle recommended for component part use and for use on panels with a .080 inch (2.03 mm) maximum thickness.
5. Wrench flats are to accommodate standard wrench opening in accordance with FED-STD-H28, appendix 10.
6. All undimensioned pictorial configurations are for reference purposes only.
7. Dimension A defines the overall length of the connector when assembled to the cable.
8. Series SMA, socket contact interface, in accordance with MIL-STD-348.
9. All threads shall conform to FED-STD-H28.

FIGURE 1. General configuration.

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TABLE I. Dash numbers, cross-reference, and dimensions.

Dash no. M39012/81-	Applicable cable 1/ M17/	Dimension A (millimeters)		Dimension B (millimeters)
		Before assembly	After assembly	
CATEGORY A – FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 2/				
3009 4009	133-RG405 *	.518 (13.16 mm) Maximum	.505 (12.83 mm) Maximum	Not applicable
3010 4010	130-RG402 *			
CATEGORY E – FIELD REPLACEABLE (STANDARD ASSEMBLY TOOL KIT) 3/ 4/				
3005 5/	133-RG405 * #	Not applicable	.850 (21.59 mm) maximum	Not applicable 6/
3006 5/	130-RG402 * #			
3007 2/ 4007 2/	133-RG405 * #			
3008 2/ 4008 2/	130-RG402 * #			
3011 2/ 4011 2/	133-RG405 * #		.550 (13.97 mm) maximum	.275 ±.010 (6.98 ±0.25 mm)
3012 2/ 4012 2/	130-RG402 * #			

See notes at end of table.

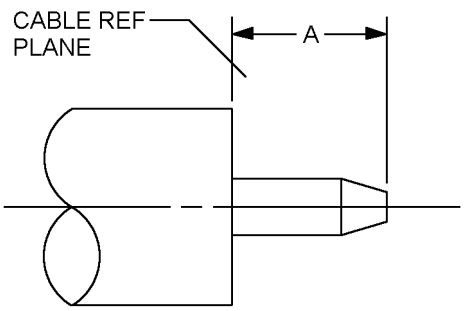
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TABLE I. Dash numbers, cross-reference, and dimensions – Continued.

Dash number M39012/81-	Applicable cable 1/ M17/	Tool number M22520/	Positioning dies M22520/	Locator pins 7/ M22520/36	Crimp dies M22520/5	Dimension A 8/ 9/		Dimension B
						Before crimping 10/	After crimping	
CATEGORY F – FIELD REPLACEABLE (MIL-DTL-22520 CRIMP TOOL) 2/								
3207 11/ 4207 11/	133-RG405 *#	36-01	36-02	-05, -17 or -22	--------	.750 (19.05 mm) maximum	.620 (15.7 mm) maximum	Not applicable 6/
3208 11/ 4208 11/	130-RG402 *#		36-03					
3210 4210	130-RG402 *#	5-01	--------	--------	-05 or -41 cavity B	.691 (17.55 mm) maximum	.691 (17.55 mm) maximum	
3213 4213	133-RG405 *#	5-01	--------	--------	-05 or -41 cavity B	.511 (12.98 mm) maximum	.511 (12.98 mm) maximum	Not applicable 6/

1/ MIL-DTL-17 cables are specified by the basic number. The latest version of each cable shall be applicable.
2/ These connectors have captivated center contacts.
3 Kit number – Omni Spectra T-200, Amphenol 901-2500, or equivalent.
4/ All corrosion resistant steel bodied connectors shall be gold plated in accordance with ASTM B488, type II, code C, class 1.27, at least in the area of solder attachment.
5/ Inactive for new design. Not for Air Force or Navy use.
6/ This dimension is furnished in the manufacturers assembly instructions.
7/ The locators required shall be indicated in the assembly instructions.
8/ Dimensions are in inches.
9/ Metric equivalents are given for information only.
10/ The A dimension before crimping determines the closeness of a bend in the cable to the rear of the connector.
11/ Not for Army use. For OEM use only.
* Cable to be used when performing test requiring cable. All cable dash numbers in accordance with MIL-DTL-17 specification sheet accommodate this connector.
\# These cables are stripped using the dimensions as specified on figure 2.

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Cable *	Dimension "A"	
	Minimum	Maximum
M17/130-RG402	.080 (2.03 mm)	.090 (2.29 mm)
M17/133-RG405	.065 (1.65 mm)	.075 (1.91 mm)

* Dimensions apply to all cables on the below specification sheets.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tapering of center conductor is optional.

FIGURE 2. Cable stripping dimensions for category E and category F connectors.

ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 18 GHz.

Voltage rating: The voltage rating shall be as specified in table II.

TABLE II. Voltage rating.

Cables	Voltage max. (sea level) V rms	Voltage max. at 70,000 ft. (4.437 kPa) V rms
M17/133-RG405 *	335	85
M17/130-RG402 *	500	125

* These voltages apply to all cable Part or Identifying Number (PIN) on these specification sheets even if not listed.

Temperature rating: -65°C to +165°C.

Category F: Connectors using semirigid cables with standard stripping dimensions and using standard military assembly tools. The method of assembly of the connector to the cable shall be solderless.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and MIL-STD-348.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds (.22 Nm), maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds (9.49 to 13.56 Nm).

Mating characteristics: See MIL-STD-348 and figure 3 for dimensions.

Center contact (socket):

Oversize test pin: .0375 +.0001 (0.952 mm +.002 mm).

Test pin finish: 16 microinches (0.406 µm).

Insertion depth: .030/.045 inch (0.76 mm/1.14 mm).

Number of insertions: 3.

Insertion force test:

Steel test pin diameter: .0370 +.0001 inch (0.940 mm +.002 mm).

Insertion depth: .050/.075 inch (1.27 mm/1.90 mm).

Test pin finish: 16 microinches (0.406 µm).

Insertion force: 2 pounds, maximum (8.90 N).

Withdrawal force test:

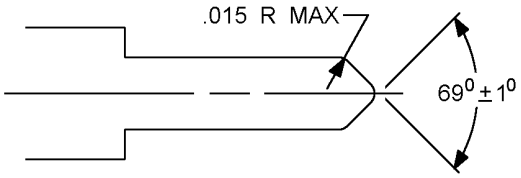
Steel test pin diameter: .0355 -.0001 inch (0.902 mm -.002 mm).

Insertion depth: .050/.075 inch (1.27 mm/1.90 mm).

Test pin finish: 16 microinches (0.406 mm).

Withdrawal force: 1 ounce, minimum (.28 N).

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Inches	mm
.015	.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 3. Test pin dimensions.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: In accordance with MIL-STD-202, method 302, test condition B, 5,000 megohms minimum.

Center contact retention:

Axial force: 6 pounds, minimum (26.69 N) axial force. Applicable to captivated center contacts only.

Radial torque: Not applicable.

Corrosion (salt spray): In accordance with MIL-STD-202 method 101, test condition B.

Voltage standing wave ratio (VSWR): From 500 MHz to 12.4 GHz, or approximately 80 percent of the upper cutoff frequency of the cable, whichever is lower.

Cable group	VSWR	
	Captive	Non-captive
M17/133-RG405*	1.07 +0.01F (F in GHz)	1.07 +0.008F (F in GHz)
M17/130-RG402*	1.05 +0.01F (F in GHz)	1.05 +0.008F (F in GHz)

* VSWR values apply to all cable PINs on these specification sheets even if not listed.

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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025 +.002F (F in GHz).

Item 16: VSWR shall be less than 1.025 +.002F (F in GHz).

Second step of VSWR checkout procedure – VSWR shall be less than 1.080 +.005F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable
Braid to body:	.5	Not applicable

NOTE: Five milliohms are permissible on passivated steel bodied connectors.

Dielectric withstanding voltage at sea level: In accordance with MIL-STD-202, method 301.

Cable group	V rms
M17/133-RG405 *	1,000
M17/130-RG402 *	1,500

* Values apply to all cable PINs on these specification sheets even if not listed.

Vibration, high frequency: In accordance with MIL-STD-202, method 204, test condition D, except the method of mounting shall be approved by the qualifying activity. No discontinuity permitted.

Shock: In accordance with MIL-STD-202, method 213, test condition I. No discontinuity permitted.

Thermal shock: In accordance with MIL-STD-202, method 107, test condition B, except test high temperature shall be +115°C.

Moisture resistance: In accordance with MIL-STD-202, method 106. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

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Corona level:

Altitude: 70,000 feet (4.437 kPa).

Cable group	Volts, min.
M17/133-RG405 *	250
M17/130-RG402 *	375

* Values apply to all cable PINs on these specification sheets even if not listed.

RF high potential withstanding voltage:

Voltage and frequency: Tested at a frequency from 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable group	Volts, min.
M17/133-RG405 *	670
M17/130-RG402 *	1,000

* Values apply to all cable PINs on these specification sheets even if not listed.

Cable retention force: The cable retention force shall be as specified in table III.

TABLE III. Cable retention force.

Cable *	Pounds, minimum (N)	Torque 1/ Inch-ounces (Nm)
M17/133-RG405	30 (133.45 N)	16 (.11 Nm)
M17/130-RG402	60 (266.90 N)	55 (.39 Nm)

* Values apply to all cable PINs on the below specification sheets even if not listed.
1/ Torque to be applied 4 inches maximum from the end of the connector. Reverse bend not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum, tested at frequency between 2 and 3 GHz..

RF insertion loss: .03 $\sqrt{F\ (GHz)}$ dB maximum tested at 9 to 12.4 GHz.

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Supplemental test requirements for qualification and group C inspection for category F connectors.

Six additional connectors shall be selected from the production lot and terminated to produce three cable leads, twelve inches minimum, sixteen inches maximum. The three assemblies shall be subjected to the following tests in sequence: VSWR, thermal shock, VSWR, and cable retention.

The following exceptions to the requirements apply to this supplemental test only:

VSWR of the cable assembly: From 0.5 to 18 GHz.

Test cable assembly	VSWR (initial)	VSWR (after thermal shock)
M17/133-RG405 *	1.15 +.023F (F in GHz)	1.15 +.028F (F in GHz)
M17/130-RG402 *	1.10 +.023F (F in GHz)	1.10 +.025F (F in GHz)

* Values apply to all cable PINs on these specification sheets even if not listed.

Thermal shock: 10 cycles.

Group qualification: See table IV.

PIN: M39012/81 (dash number in table I or “B” number in table V).

PIN cross-reference: See table VI.

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TABLE IV. Group qualification and retention testing. 1/

Group	Submission and qualification of any of the following dash numbers 2/ M39012/	Qualifies the following dash numbers M39012/
I	81-*02 82-*02 83-*02	81-*01 81-*02 82-*01 82-*02 83-*01 83-*02
II	81-*04 82-*04 83-*04	81-*03 81-*04 82-*03 82-*04 83-*03 83-*04
III	81-*06 82-*06 83-*06	81-*05 81-*06 82-*05 82-*06 83-*05 83-*06
IV	81-*08 82-*08 83-*08	81-*07 81-*08 82-*07 82-*08 83-*07 83-*08
V	81-*10 82-*10 83-*10	81-*09 81-*10 82-*09 82-*10 83-*09 83-*10
VI	81-*208 82-*208 83-*208	81-*207 81-*208 82-*207 82-*208 83-*207 83-*208
VII	81-*210 82-*210 83-*210	81-*210 81-*213 82-*210 82-*215 82-*216 83-*210 83-*213

See notes at end of table.

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TABLE IV. Group qualification and retention testing – Continued. 1/

1/ If a connector manufacturer produces a connector which meets all the requirements for two or more connector PINs (within the same series), the manufacturer may receive qualification approval for two or more connector PINs by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate PIN. For group qualification, the connectors must be of similar design.

2/ For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right-hand column. The part does not necessarily have to be the part initially qualified. Qualification of connectors qualifies connectors of the same body material and finish only.

* Denotes body material and connector requirement.

TABLE V. Category B – Non-field replaceable (special tools may be required). 1/

NOT FOR ARMY, NAVY, OR AIR FORCE USE. FOR OEM USE

M39012/81B 2/	Applicable cable 3/	Dimension A Maximum 6/ 7/
3001 4/	M17/133-RG405 *	.850 (21.59 mm)
3002 4/	M17/130-RG402 *	
3003 5/ 4003 5/	M17/133-RG405 *	
3004 5/ 4004 5/	M17/130-RG402 *	

1/ Dimension B shown on figure 1 is not applicable to these connectors.
2/ All corrosion resistant steel bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold plated in accordance with ASTM B488, type II, code C, class 1.27.
3/ MIL-DTL-17 cables are specified by the basic number. The latest version of each cable shall be applicable.
4/ In active for new design.
5/ These connectors have captivated center contacts.
6/ Dimensions are in inches.
7/ Metric equivalents are given for information only.
* Cable to be used when performing tests requiring cable. All cable PIN's on this specification sheet shall accommodate this connector.

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TABLE VI. Cross reference of PIN's.

Replacement PIN 1/ M39012/81	Substitute for PIN or type designation M39012/81
B3001	-3001
B3002	-3002
B3003	-3801, -3803, -3003
B3004	-3802, -3804, -3004
B4003	-4801, -4803, -4001, -4003
B4004	-4802, -4804, -4002, -4004
-3007	-3805, -3807
-3008	-3806, -3808
-4007	-4805, -4807
-4008	-4806, -4808

1/ The connectors that are in stock or distribution that were previously marked with the old PIN shall also be considered acceptable for Government use until connector stock is purged. Applies to category B PIN change only. (i.e., M39012/81-3001 to M39012/81B3001).

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-39012, this document references the following:

MIL-DTL-17
MIL-DTL-22520
MIL-STD-348
MIL-STD-202
FED-STD-H28
ASTM B488

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2006-010)

Review activities:
Army - AM, AT, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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